

Bohr atomic model

In 1913, Bohr suggested that, *positively charged nucleus is at the centre and negatively charged electrons revolve round the nucleus in various circular orbit*. The electrons revolving round the nucleus only in certain permitted orbits are called *energy levels*. Each energy level has a certain *fixed amount of energy*. The larger the orbit (*i.e.*, larger radius), the greater is the energy of electrons. Electrons give out energy in the form of electromagnetic radiation when it jumps from *higher to lower level*.

Postulates of Bohr atomic model

Bohr atomic model is based on the following postulates:

- An atom consists of positively charged nucleus at the centre.
- The negatively charged electrons move round the nucleus in various orbits known as *stationary energy levels*. The electrons cannot emit radiations when moving in their own stationary level.
- The Coulombian and Newtonian forces are applicable in the domain of the atom.
- The electrons revolve round the nucleus in various circular orbits and the angular momentum
- $mvr = \frac{nh}{2\pi}$
- where $n (= 1, 2, 3, 4, \dots)$ is called the quantum number, and h is the Planck's constant.
- When an electron jumps from a higher energy level to a lower energy level, it gives out electromagnetic radiations of a particular frequency

$$\Delta E = E_2 - E_1 = h\nu$$

where E_2 is the energy of the upper level and E_1 is the energy of the lower level and ν is the frequency of the electromagnetic radiation.

Radius of and total energy of atom on the basis of Bohr atomic model

Let us consider the motion of an electron of mass ' m ' and charge ' e ' moving uniformly with speed ' v ', in a circular orbit of radius ' r ' around a nucleus of charge Ze . Then it experiences an acceleration v^2/r towards the centre of the circle.

The electrostatic force of attraction acts between the nucleus and the revolving electron and tend to attract the electron towards the nucleus. According to Coulomb's law this force is given by

$$F_e = \frac{1}{4\pi\epsilon_0} \frac{Ze \cdot e}{r^2}$$

$$F_e = \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2} \quad (1)$$

The centripetal force $F_c = \frac{mv^2}{r}$ is holding the electron in the orbit from the nucleus is provided by the electrostatic force F_e between them.

Bohr's postulates states that, the electrostatic force of attraction provides necessary centripetal force for revolution of electron in the orbit, *i.e.*

$$\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2} = \frac{mv^2}{r}$$

$$v^2 = \frac{Ze^2}{4\pi\epsilon_0 mr} \quad (2a)$$

$$\therefore v = \sqrt{\frac{Ze^2}{4\pi\epsilon_0 mr}} \quad (2b)$$

Again according to the Bohr postulate, the orbital angular momentum of the revolving electron must be equal to the integer multiple of $\frac{h}{2\pi}$ i.e.,

$$mvr = \frac{nh}{2\pi}$$

$$\therefore v = \frac{nh}{2\pi mr} \quad (3)$$

where $n (= 1, 2, 3, 4, \dots)$

Equating equations (2a) and (3) we get

$$\frac{nh}{2\pi mr} = \sqrt{\frac{Ze^2}{4\pi\epsilon_0 mr}}$$

Squaring

$$\frac{n^2 h^2}{4\pi^2 m^2 r^2} = \frac{Ze^2}{4\pi\epsilon_0 mr}$$

$$r = \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2}$$

$$\therefore r_n = \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2} \quad (4)$$

Where, r_n is the radius of the n^{th} orbital electron.

For Hydrogen atom $Z = 1$. Thus, the radius of the n^{th} orbit of Hydrogen atom is

$$\therefore r_n = \frac{n^2 h^2 \epsilon_0}{\pi m e^2} \quad (5)$$

Now substituting the value of r from equation (4) in equation (3) we get

$$v_n = \frac{nh}{2\pi m \times \frac{n^2 h^2 \epsilon_0}{\pi Z m e^2}}$$

$$\therefore v_n = \frac{Ze^2}{2\epsilon_0 nh} \quad (6)$$

Where v_n is the velocity of the n^{th} orbital electron.

The kinetic energy of the revolving electron is

$$KE = \frac{1}{2} m v^2 = \frac{1}{2} m \times \frac{Ze^2}{4\pi\epsilon_0 mr}$$

$$\therefore KE = \frac{Ze^2}{8\pi\epsilon_0 r} \quad (7)$$

and the potential energy of the electron is-

$$PE = -\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r} \quad (8)$$

The total energy E of the revolving electron in a stable orbit is given by

$$\begin{aligned}
 E &= KE + PE \\
 E &= \frac{Ze^2}{8\pi\epsilon_0 r} - \frac{Ze^2}{4\pi\epsilon_0 r} \\
 \therefore E &= -\frac{Ze^2}{8\pi\epsilon_0 r} \quad (9)
 \end{aligned}$$

Now substituting the value of r from equation (4) in equation (9) we get

$$\begin{aligned}
 E_n &= -\frac{Ze^2}{8\pi\epsilon_0 \times \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2}} \\
 \therefore E_n &= -\frac{mZ^2 e^4}{8\epsilon_0^2 n^2 h^2} \quad (10)
 \end{aligned}$$

Where E_n is the energy of the n^{th} orbital electron.

For Hydrogen atom $Z = 1$. Thus, the energy of the n^{th} orbit of Hydrogen atom is

$$E_n = -\frac{me^4}{8\epsilon_0^2 n^2 h^2} \quad (11)$$

For $n = 1$, we can write equation (11) as

$$E_1 = -\frac{me^4}{8\epsilon_0^2 h^2} \quad (12)$$

From equation (11) and (12) we can write

$$E_n = \frac{E_1}{n^2}$$

When an electron jumps from upper energy level n_2 to lower energy level n_1 , we can write

$$\Delta E = E_{n_2} - E_{n_1} \quad (13)$$

From equation (10) we can write

$$E_{n_2} = -\frac{mZ^2 e^4}{8\epsilon_0^2 n_2^2 h^2} \quad (14)$$

and

$$E_{n_1} = -\frac{mZ^2 e^4}{8\epsilon_0^2 n_1^2 h^2} \quad (15)$$

From equation (13), (14) and (15) we can write

$$\begin{aligned}
 \Delta E &= -\frac{mZ^2 e^4}{8\epsilon_0^2 n_2^2 h^2} + \frac{mZ^2 e^4}{8\epsilon_0^2 n_1^2 h^2} \\
 \Delta E &= \frac{mZ^2 e^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (16)
 \end{aligned}$$

Where, ΔE is the energy of the emitted photon.

For Hydrogen atom $Z = 1$. Thus, the energy of the Hydrogen atom in this case is

$$\Delta E = \frac{me^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (17)$$

But, we know, $\Delta E = h\nu$. So that we can write equation (17) as

$$\nu = \frac{me^4}{8\varepsilon_0^2 h^3} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (18)$$

But

$$c = \nu\lambda, \quad \therefore \quad \nu = \frac{c}{\lambda}$$

So that we can write equation (18) as

$$\frac{c}{\lambda} = \frac{me^4}{8\varepsilon_0^2 h^3} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda} = \frac{me^4}{8\varepsilon_0^2 ch^3} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$\frac{1}{\lambda} = \bar{\nu}$ is called the wave number.

$$\bar{\nu} = \frac{me^4}{8\varepsilon_0^2 ch^3} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$\frac{me^4}{8\varepsilon_0^2 ch^3} = R$ and is known as Rydberg constant.

$$\therefore \quad \bar{\nu} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (19)$$

Spectral Series of Hydrogen Atom

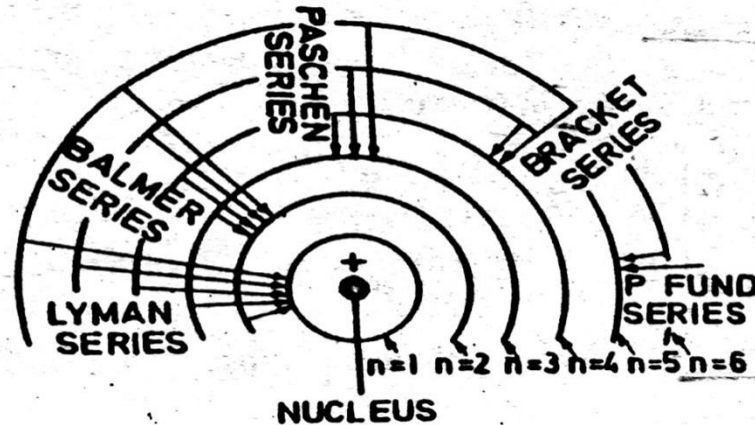


Fig.1. Spectral Series of Hydrogen

We know, the wave number of the Hydrogen atom on the basis of Bohr atomic model

$$\bar{\nu} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Lyman Series: When an electron jumps from the outer orbits to the first orbit, this corresponds to Lyman series.

Here, $n_1 = 1, \quad n_2 = 2, 3, 4 \dots \dots$

$$\therefore \quad \bar{\nu}_1 = R \left(\frac{1}{1} - \frac{1}{2^2} \right) = \frac{3}{4} R$$

$$\therefore \bar{\nu}_2 = R \left(\frac{1}{1} - \frac{1}{3^2} \right) = \frac{8}{9} R \quad \text{etc}$$

Balmer Series: When an electron jumps from the outer orbits to the second orbit, this corresponds to Balmer series.

Here, $n_1 = 2, \quad n_2 = 3, 4, 5 \dots \dots$

$$\therefore \bar{\nu}_1 = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{5}{36} R$$

$$\therefore \bar{\nu}_2 = R \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = \frac{3}{16} R \quad \text{etc}$$

Paschen Series: When an electron jumps from the outer orbits to the third orbit, this corresponds to Paschen series.

Here, $n_1 = 3, \quad n_2 = 4, 5, 6 \dots \dots$

$$\therefore \bar{\nu}_1 = R \left(\frac{1}{3^2} - \frac{1}{4^2} \right) = \frac{9}{144} R$$

$$\therefore \bar{\nu}_2 = R \left(\frac{1}{3^2} - \frac{1}{5^2} \right) = \frac{16}{225} R \quad \text{etc}$$

Bracket Series: When an electron jumps from the outer orbits to the fourth orbit, this corresponds to Bracket series.

Here, $n_1 = 4, \quad n_2 = 5, 6, 7 \dots \dots$

$$\therefore \bar{\nu}_1 = R \left(\frac{1}{4^2} - \frac{1}{5^2} \right) = \frac{9}{400} R$$

$$\therefore \bar{\nu}_2 = R \left(\frac{1}{4^2} - \frac{1}{6^2} \right) = \frac{5}{144} R \quad \text{etc}$$

Pfund Series: When an electron jumps from the outer orbits to the fifth orbit, this corresponds to Pfund series.

Here, $n_1 = 5, \quad n_2 = 6, 7, 8 \dots \dots$

$$\therefore \bar{\nu}_1 = R \left(\frac{1}{5^2} - \frac{1}{6^2} \right) = \frac{11}{900} R$$

$$\therefore \bar{\nu}_2 = R \left(\frac{1}{5^2} - \frac{1}{7^2} \right) = \frac{24}{1225} R \quad \text{etc}$$

Mathematical Problems**Problem-1:** Calculate the radius of the first Bohr orbit for H and He atoms.**Solution:** We know,

$$r = \frac{\epsilon_0 n^2 h^2}{\pi m Z e^2}$$

For Hydrogen atom $Z = 1$

$$r = \frac{\epsilon_0 n^2 h^2}{\pi m e^2}$$

$$r = \frac{8.854 \times 10^{-12} \times (1)^2 \times (6.63 \times 10^{-34})^2}{3.1416 \times 9.1 \times 10^{-31} \times (1.6 \times 10^{-19})^2}$$

$$r = 5.29 \times 10^{-11} m$$

$$r = 0.530 \text{ \AA}^0 \text{ (Ans)}$$

For Helium atom $Z = 2$

$$r = \frac{\epsilon_0 n^2 h^2}{2 \pi m e^2}$$

$$r = \frac{8.854 \times 10^{-12} \times (1)^2 \times (6.63 \times 10^{-34})^2}{2 \times 3.1416 \times 9.1 \times 10^{-31} \times (1.6 \times 10^{-19})^2}$$

$$r = 2.65 \times 10^{-11} m$$

$$r = 0.265 \text{ \AA}^0 \text{ (Ans)}$$

Here,

$$h = 6.63 \times 10^{-34} \text{ JS}$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

Problem-2: Show that the velocity of an electron in the 1st orbit in hydrogen atom is $\left(\frac{1}{137}\right)c$, where 'c' is the velocity of light.**Solution:** We know,

$$v = \frac{Z e^2}{2 \epsilon_0 n h}$$

For Hydrogen atom $Z = 1$

$$v = \frac{e^2}{2 \epsilon_0 n h}$$

$$\frac{v}{c} = \frac{e^2}{2 \epsilon_0 n h c}$$

$$\frac{v}{c} = \frac{(1.6 \times 10^{-19})^2}{2 \times 8.854 \times 10^{-12} \times 1 \times 6.63 \times 10^{-34} \times 3 \times 10^8}$$

$$\frac{v}{c} = \frac{1}{137}$$

$$\therefore v = \left(\frac{1}{137}\right)c$$

(Showed)

Here,

$$h = 6.63 \times 10^{-34} \text{ JS}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

Problem-3: Calculate the radius of the third orbit of hydrogen atom. Also calculate the energy of electrons in that orbit. (Given $h = 6.62 \times 10^{-34} \text{ JS}$, $m = 9.1 \times 10^{-31} \text{ kg}$, $e = 1.6 \times 10^{-19} \text{ C}$)

Solution: We know, for Hydrogen atom

$$r = \frac{\epsilon_0 n^2 h^2}{\pi m e^2}$$

$$r = \frac{8.854 \times 10^{-12} \times (3)^2 \times (6.63 \times 10^{-34})^2}{3.1416 \times 9.1 \times 10^{-31} \times (1.6 \times 10^{-19})^2}$$

$$r = 4.78 \times 10^{-10} \text{ m}$$

$$r = 4.78 \text{ \AA}^0 \text{ (Ans)}$$

Energy of an atom is

$$E = -\frac{me^4 Z^2}{8\epsilon_0^2 n^2 h^2}$$

For Hydrogen atom $Z = 1$

$$E = -\frac{me^4}{8\epsilon_0^2 n^2 h^2}$$

$$E = -\frac{9.1 \times 10^{-31} \times (1.6 \times 10^{-19})^4}{8 \times (8.854 \times 10^{-12})^2 \times (3)^2 \times (6.63 \times 10^{-34})^2}$$

$$E = -2.41 \times 10^{-19} \text{ J}$$

$$E = -\frac{2.41 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV}$$

$$E = -1.506 \text{ eV (Ans)}$$

Problem-4: How much energy is required to remove an electron from $n=3$ or $n=2$ state in Hydrogen atom.

Solution: We know, Energy of an atom is

$$E = \frac{me^4 Z^2}{8\epsilon_0^2 h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

For Hydrogen atom $Z = 1$

$$E = \frac{me^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$E = \frac{9.1 \times 10^{-31} \times (1.6 \times 10^{-19})^4}{8 \times (8.854 \times 10^{-12})^2 \times (6.63 \times 10^{-34})^2} \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$$

$$E = 2.16 \times 10^{-18} \text{ J}$$

$$E = \frac{2.16 \times 10^{-18}}{1.6 \times 10^{-19}} \text{ eV}$$

$$E = 13.5 \text{ eV (Ans)}$$

Here,

$$h = 6.63 \times 10^{-34} \text{ JS}$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

Here,

$$h = 6.63 \times 10^{-34} \text{ JS}$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

Problem-5: Calculate the energy required to excite the hydrogen atom from the ground state ($n=1$) to the first excited state ($n=2$).

Solution: We know, Energy of an atom is

$$E = \frac{me^4 Z^2}{8\varepsilon_0^2 h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

For Hydrogen atom $Z = 1$

$$E = \frac{me^4}{8\varepsilon_0^2 h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$E = \frac{9.1 \times 10^{-31} \times (1.6 \times 10^{-19})^4}{8 \times (8.854 \times 10^{-12})^2 \times (6.63 \times 10^{-34})^2} \left(\frac{1}{1^2} - \frac{1}{2^2} \right)$$

$$E = 1.62 \times 10^{-18} \text{ J}$$

$$E = \frac{1.62 \times 10^{-18}}{1.6 \times 10^{-19}} \text{ eV}$$

$$E = 10.12 \text{ eV (Ans)}$$

Here,

$$h = 6.63 \times 10^{-34} \text{ JS}$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\varepsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N} - \text{m}^2$$

Problem-6: The wavelength of second line of Balmer series in the hydrogen spectrum is 4861 Å . Calculate the wavelength of first line.

Solution: We know,

$$\bar{\nu} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

For the second line

$$\bar{\nu}_2 = R \left[\frac{1}{2^2} - \frac{1}{4^2} \right]$$

$$\bar{\nu}_2 = \frac{3}{16} R$$

$$\frac{1}{\lambda_2} = \frac{3}{16} R \quad (1)$$

For the first line

$$\bar{\nu}_1 = R \left[\frac{1}{2^2} - \frac{1}{3^2} \right]$$

$$\bar{\nu}_1 = \frac{5}{36} R$$

$$\frac{1}{\lambda_1} = \frac{5}{36} R \quad (2)$$

Dividing equation (2) by equation (1)

$$\frac{\lambda_1}{\lambda_2} = \frac{36}{5} \times \frac{3}{16} \Leftrightarrow \frac{\lambda_1}{\lambda_2} = \frac{27}{20}$$

$$\lambda_1 = \frac{27}{20} \times \lambda_2 \Leftrightarrow \lambda_1 = \frac{27}{20} \times 4861 \quad \therefore \quad \lambda_1 = 6563 \text{ Å (Ans)}$$